

# Import / Export: User-Reviewed Next Step Suggestions

A real-world concept brief for turning imported railway data into accountable operational decisions.

AI recommends. A railway professional decides. Every decision is traceable.

## Hackathon opportunity

RailMind already parses CSV and JSON event data, validates records, calculates passenger and delay impact, and generates next-step suggestions. The highest-value improvement is to make those suggestions operationally useful without allowing the AI to act unchecked.

The proposed experience is RailMind Control Handoff: a reviewed handoff from imported evidence to a human-approved railway action.

## The flagship workflow

|              |                                                                          |
|--------------|--------------------------------------------------------------------------|
| 1. Import    | Upload an incident, maintenance, or service-disruption file.             |
| 2. Check     | Detect invalid records, duplicate events, and conflicts between sources. |
| 3. Recommend | Rank a next action with evidence, confidence, impact, and missing data.  |
| 4. Review    | A supervisor approves, rejects, or asks for more information.            |
| 5. Handoff   | Create a draft task, simulation, escalation, or role-specific export.    |
| 6. Replay    | Preserve the recommendation, decision, outcome, and timestamp for audit. |

# Ten distinctive real-world changes

## 1. Import-to-Action Safety Passport

Give each uploaded file a data-quality score, affected assets, passenger impact, recommended action, responsible team, approval state, and decision history.

## 2. What Changes If We Wait?

Compare acting now, waiting 30 minutes, and doing nothing. Show expected delay, passenger exposure, and service disruption for each option.

## 3. Cross-Source Conflict Detector

Identify contradictory station, train, severity, timestamp, or asset information before an import reaches operations.

## 4. Approval Levels

Route recommendations through realistic governance: informational, supervisor approval, multi-team approval, or incident-commander approval.

## 5. Import a Situation, Export a Briefing

Produce a station-manager brief, maintenance work order, control-room summary, passenger communication draft, or executive report.

## 6. Decision Replay

Show the chain from imported evidence to recommendation, human decision, and result so an incident can be reviewed later.

## 7. Similar Past Incident

Compare the imported situation with previous incidents and cite the response that worked before, such as inspecting a specific relay cabinet.

## 8. Confidence With Missing Evidence

Lower confidence when sensor readings or timestamps are absent, then suggest the next data source to import.

## 9. Data-to-Dispatch Handoff

After approval, create a structured draft task with team, location, priority, equipment, deadline, and escalation rule.

## 10. Reversible Approval

Clearly label which actions can be cancelled, rerun, or reversed, and which require a formal escalation to undo.

# Suggested demonstration scenario

Upload a CSV representing a morning disruption at Howrah Junction. The data contains a critical track defect, a high-severity signal failure, and platform overcrowding during rush hour.

```
event_type,severity,description,affected_passengers,estimated_delay_minutes,priority,station_id
track_defect,critical,Major rail fracture near Howrah Junction,2500,120,9,1
signal_failure,high,Signal failure affecting eastbound services,800,45,8,1
platform_overcrowding,medium,Platform 3 overcrowded during rush hour,1200,10,5,1
```

# Expected recommendation panel

| Recommendation               | Evidence and user choice                                                                                                                                   |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Simulate delay scenarios     | 4,500 passengers are affected by 175 minutes of estimated delay.<br>Confidence: 80%. Approve, reject, or inspect details.                                  |
| Review emergency response    | A critical track defect requires a supervisor decision before a response draft is created. The import process must not silently duplicate agent execution. |
| Export control-room briefing | Generate a concise CSV, JSON, or role-specific briefing for offline review and sharing.                                                                    |

# Implementation priority

For the MVP, implement approval states, a confirmation step, executable navigation/export actions, a simulation handoff, and an audit record. Treat agent actions already triggered during confirmation as Already completed, or require an explicit rerun confirmation to prevent duplicate processing.

Success metric: a judge can upload one realistic file, understand why RailMind made a recommendation, approve one action, reject another, and inspect the recorded decision trail.